Centre for
Data Analytics

Insight

Ontologising the Health Level Seven (HL7) Standard

Dr. Ratnesh Sahay
Semantics in eHealth & Life Sciences (SeLS)
Insight Centre for Data Analytics
NUI Galway, Ireland

Semantic Web Application and Tools 4 Life Science (SWAT4LS)
Freie Universitaet Berlin
Germany
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DCU NUI Galway OÉ Gaillimh UCC University College Cork, Ireland Coláiste na hOllscoile Corcaigh UCD DUBLIN sfi Science Foundation Ireland

HL7 Ontologies

- Plug & Play Electronic Patient Records (PPEPR)
  - Funding: Enterprise Ireland
  - 2006-2009
  - 2014: PPEPR-2
  - http://www.ppepr.org/
  - Lead by me



- HL7 OWL
  - Supported by HL7
  - 2013 - ongoing
  - http://gforge.hl7.org/gf/project/hl7owl/
  - Lead by Lloyd McKenzie



Tutorial Overview

- Background
  - Ontology
  - Healthcare Interoperability

- Health Level Seven (HL7) Messaging Environment
- Plug and Play Electronic Patients Records (PPEPR)
- Aligning HL7 Ontologies
- Context & Modularity for HL7 ontologies

Ontology ?

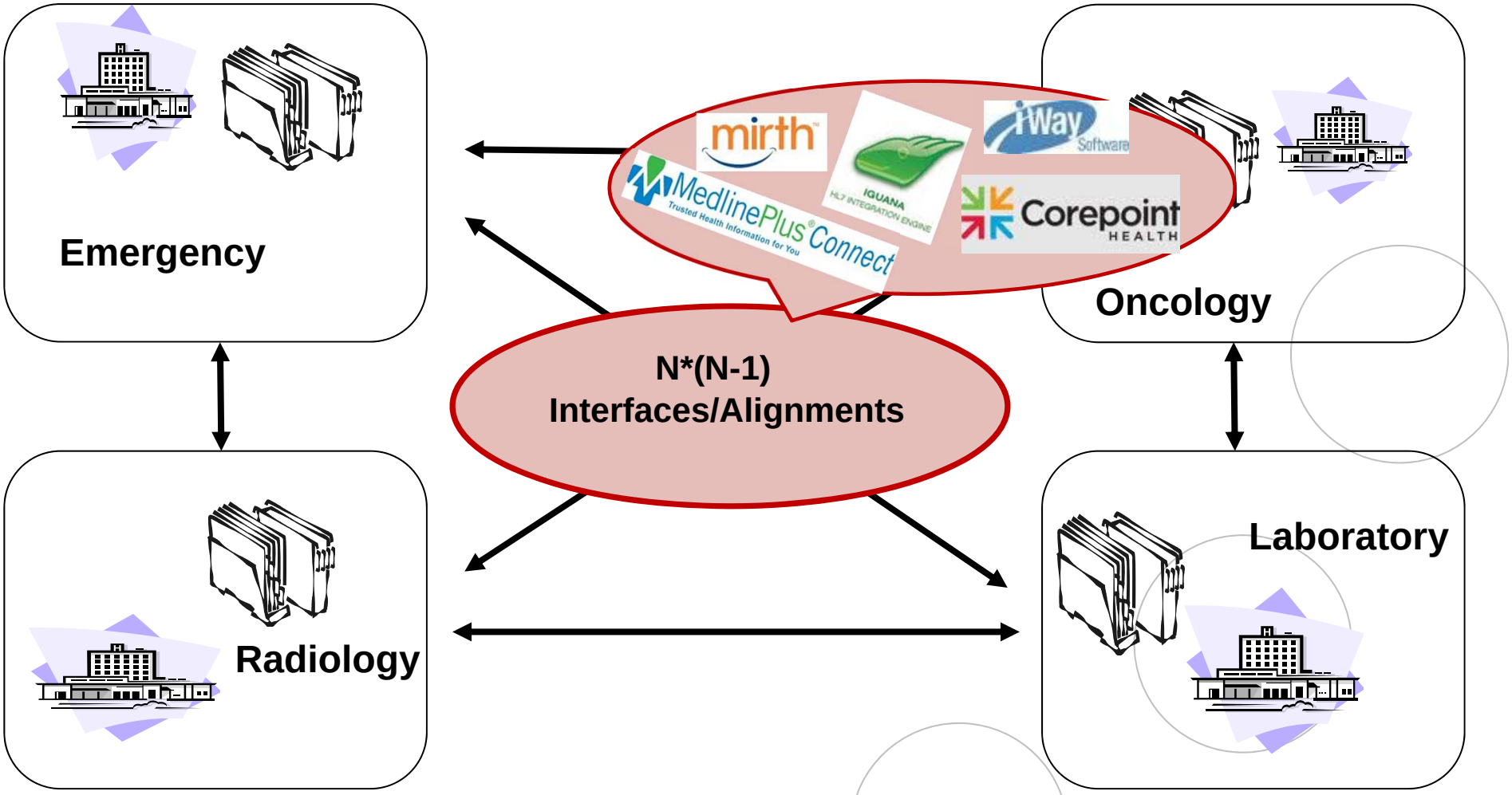
- Humans like to **classify** things !
  - Galaxies, Molecules, Genomics, Education
- The Latin term ***ontologia*** was first invented in 1613 by two German philosophers
  - Rudolf Gockel
  - Jacob Lorhard
- In context of **knowledge base systems** – Tom Gruber (Siri inventor !)
  - Toward Principles for the **Design of Ontologies** Used for Knowledge Sharing (1993)
  - A Translation Approach to Portable **Ontology Specifications** (1995)

- Ontologies are
  - *„**Explicit** Specification of a **conceptualisation**.“* — *Tom Gruber, 1993*
  - **Agreed** between groups with **explicit semantics**. — OWL Semantics, W3C, 2004
  - **Monotonic** and make Open World Assumption (OWA). — OWL Semantics, W3C, 2004
  - Good at Description of **Reality and their mappings**.

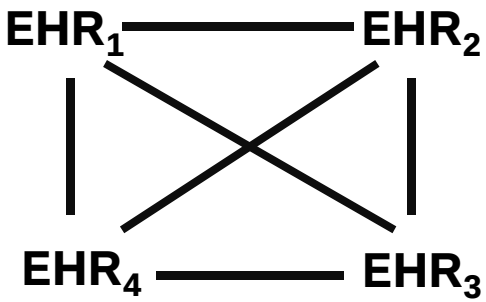
Healthcare Interoperability: Background

- 1986: **IEEE P1157 Medical Data Interchange (MEDIX)** committee introduced the concept of a **common healthcare data model**
- 1987: **HL7 Version 2**
- 1995-2005: **HL7 Version 3**
- **MEDIX** work is the core of current healthcare standards (**Health Level Seven (HL7)**, ***open*EHR**, **CEN 13606**)
- **Health Level Seven (HL7)** is the most widely deployed healthcare standard !
- 2000 onwards: HL7 **Integration platforms**
  - End-to-End bidirectional interface development (Mirth, iWay, iNTERFACEWARE)
  - Very few exit for Version 3 applications
  - None provided interoperability between Version 2 and Version 3 applications
- 2004 onwards: **Semantic Interoperability (Ontologies) for Healthcare**
  - Projects: Artemis, RIDE, SemanticHEALTH, SAPHIRE, ACGT, W3C HCLS, etc.
  - **Plug and Play Electronic Patient Record (PPEPR)** started end of 2006
  - **Healthcare Vision**: an Unified Electronic Healthcare Records (EHRs)

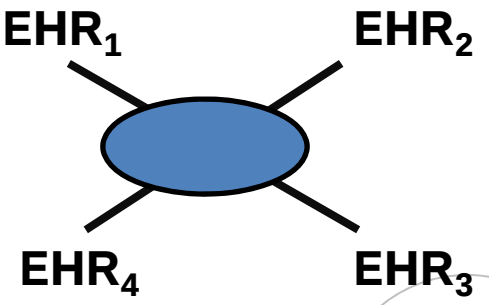
Healthcare Interoperability: Current Situation



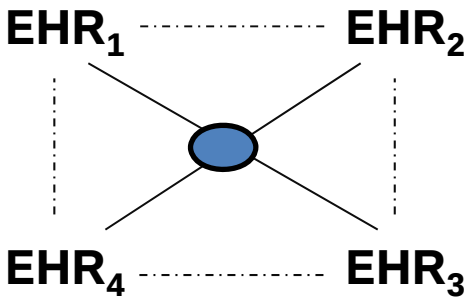
Ontological Approaches



(1) current situation
(2) local alignment = $(n \times (n-1))$

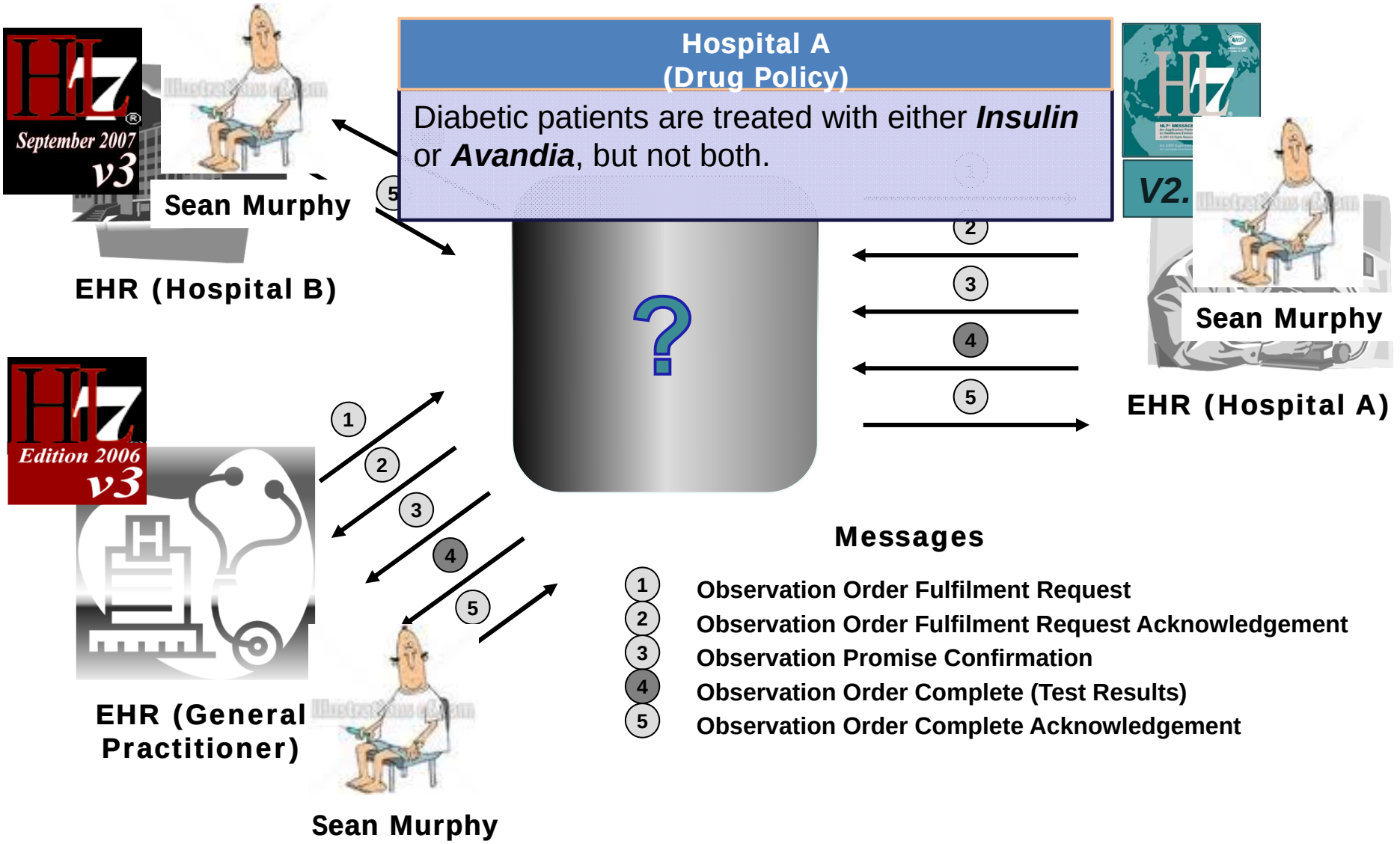


(1) ideal situation
(2) global alignment

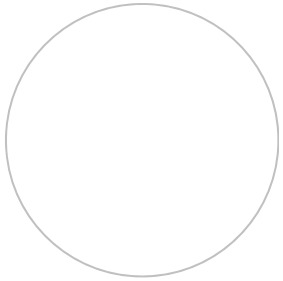


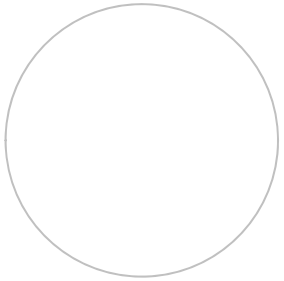
(1) Hybrid approach
(2) global and local alignments

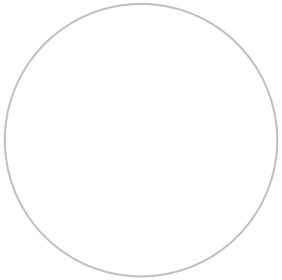
Example Scenario



Health Level Seven Standard (HL7)

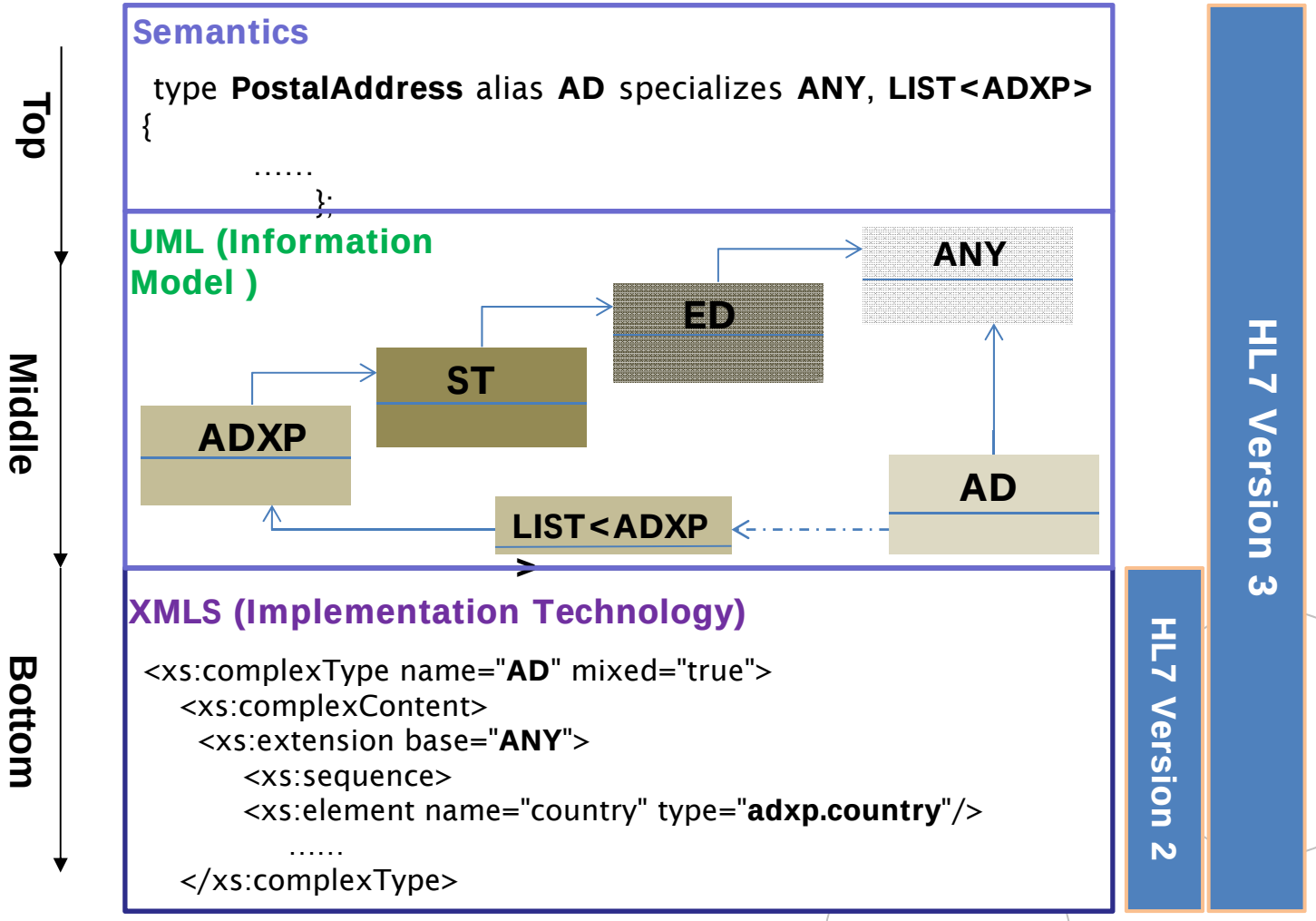






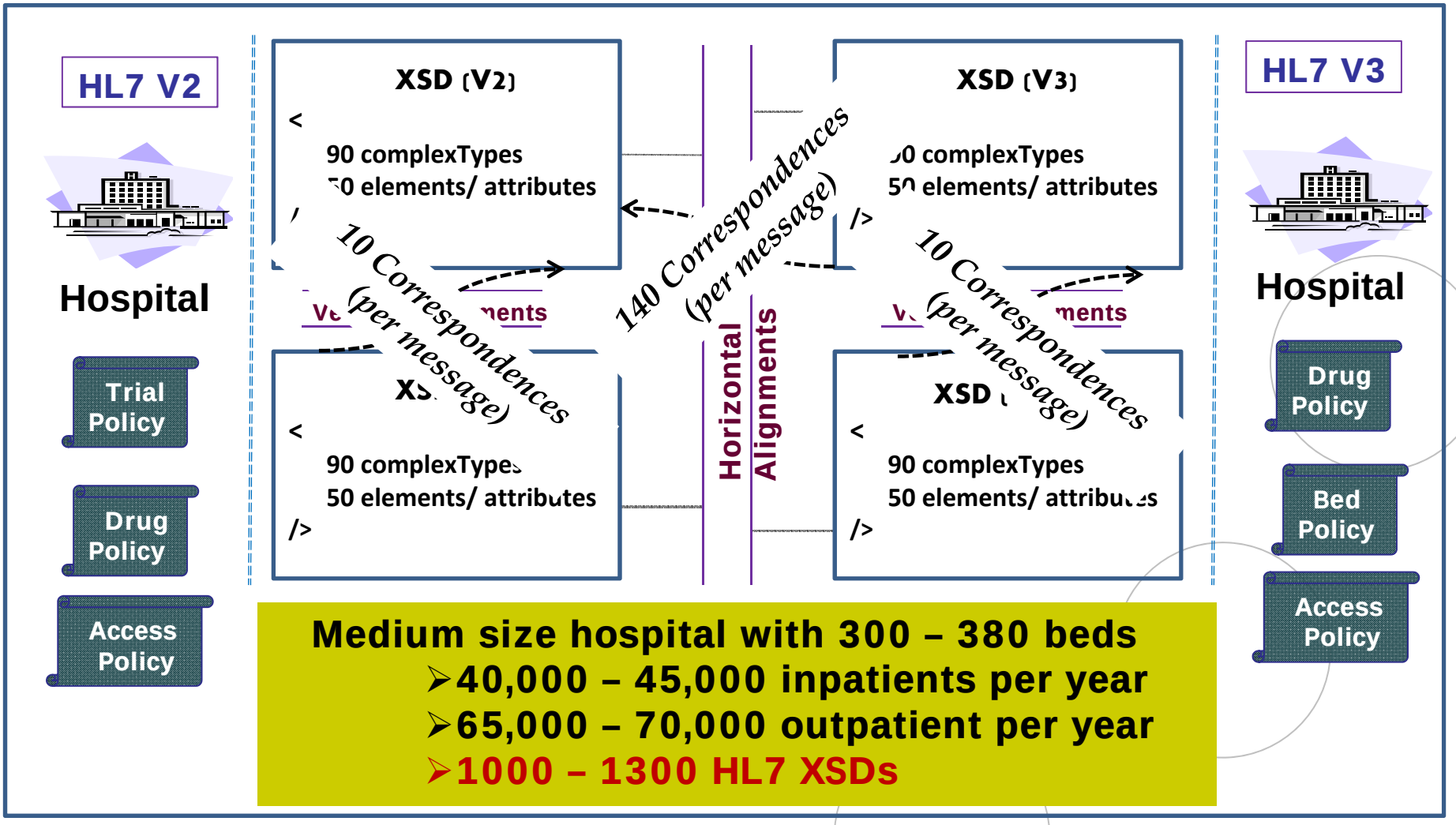
HL7 Messaging Environment - 1:

Semantics to Implementation

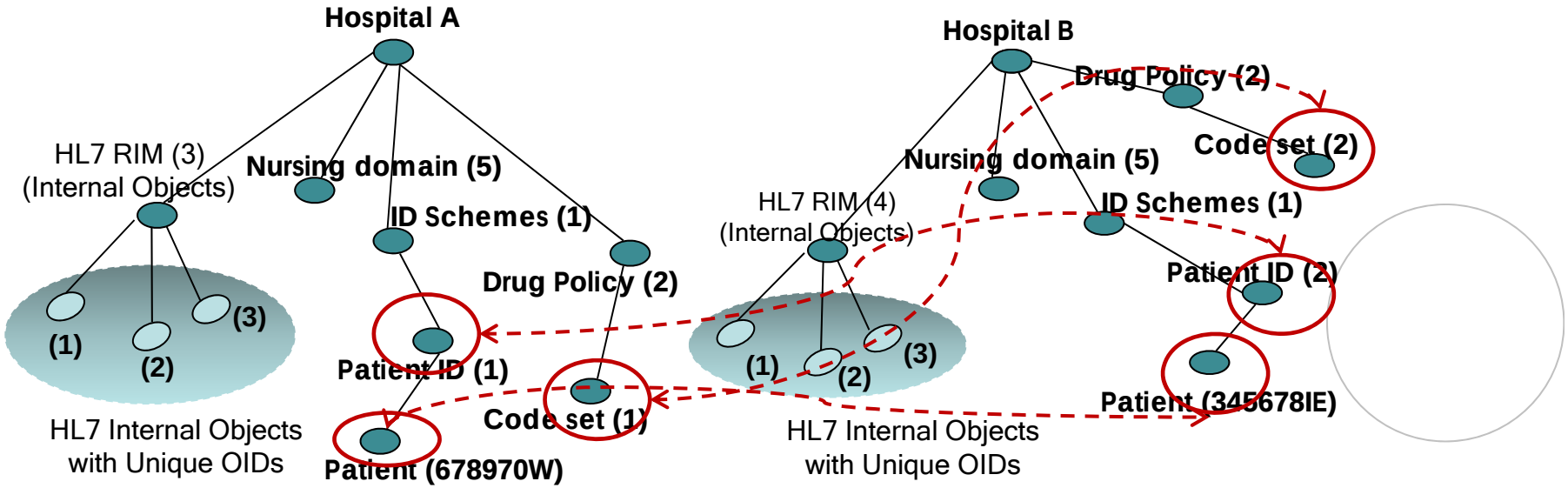


Health Level Seven (HL7) Messaging Environment : - 2

Schema, Alignment, and Local Policies

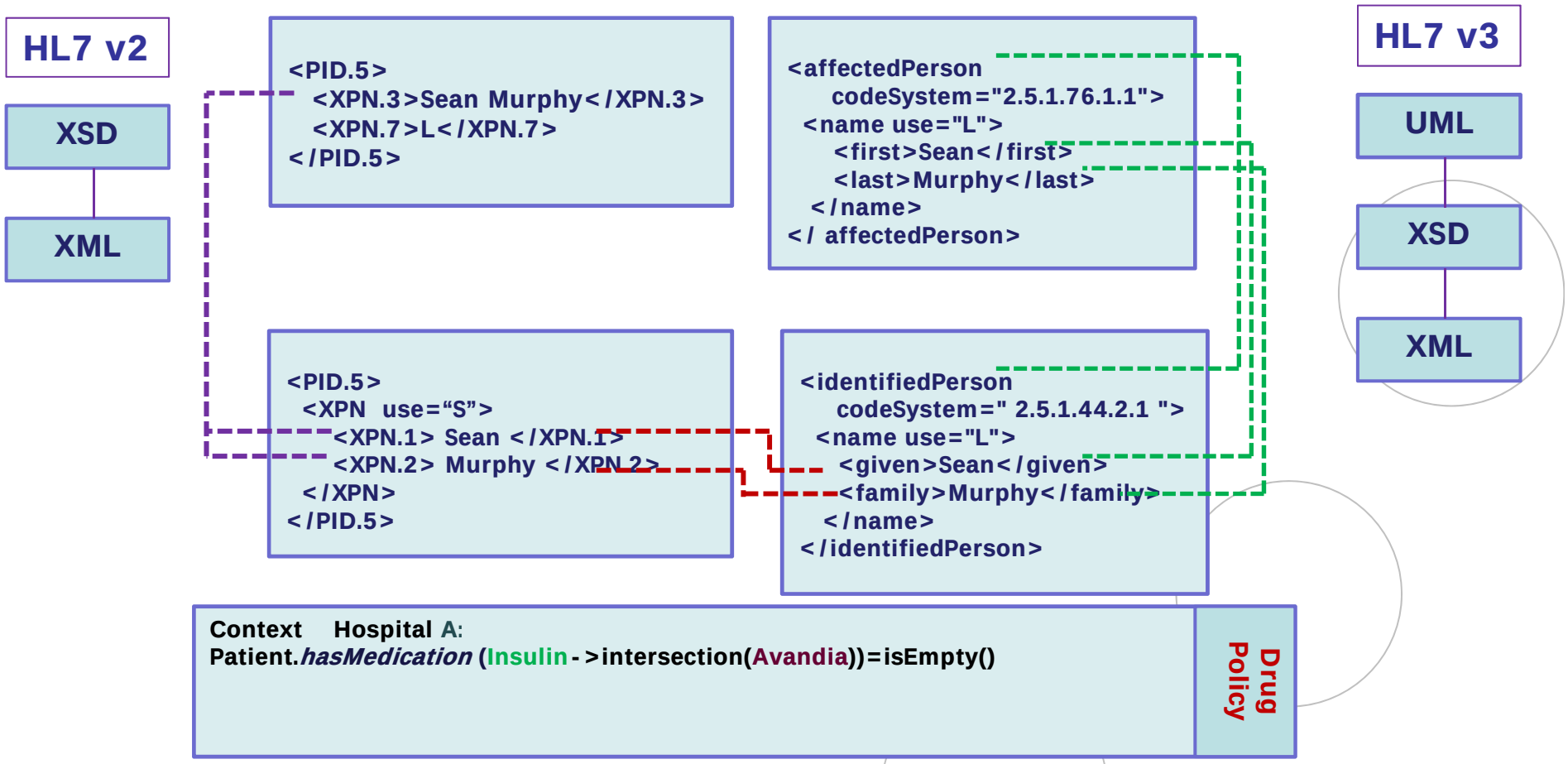


HL7 Messaging Environment – 3:
Contextual/Modular Information Structure

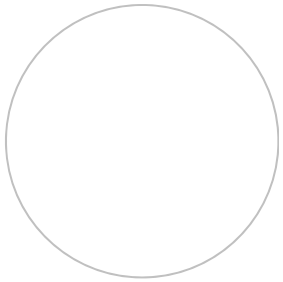


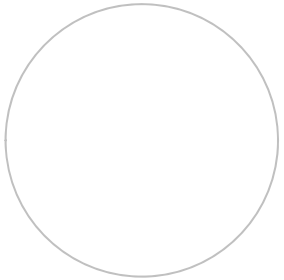
- Each entity is identified by an unique Object Identifiers (OIDs)
- Health records are arranged in separate modules
- Constraints or Policies are identifiable local modules

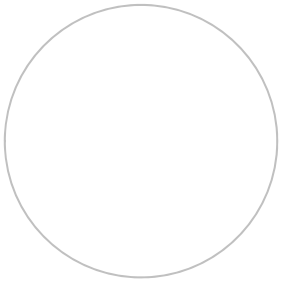
HL7 Messaging Environment – 4:
Example



Ontologising Health Level Seven Standard (HL7)





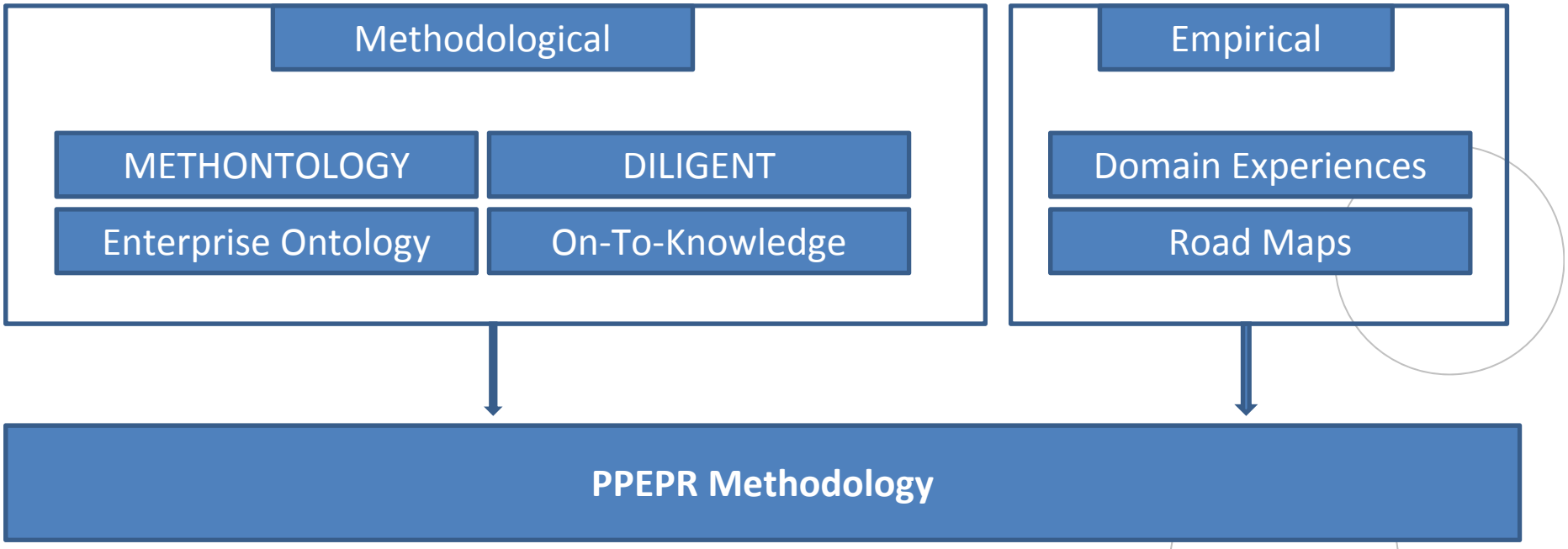


Ontology Building Methodologies

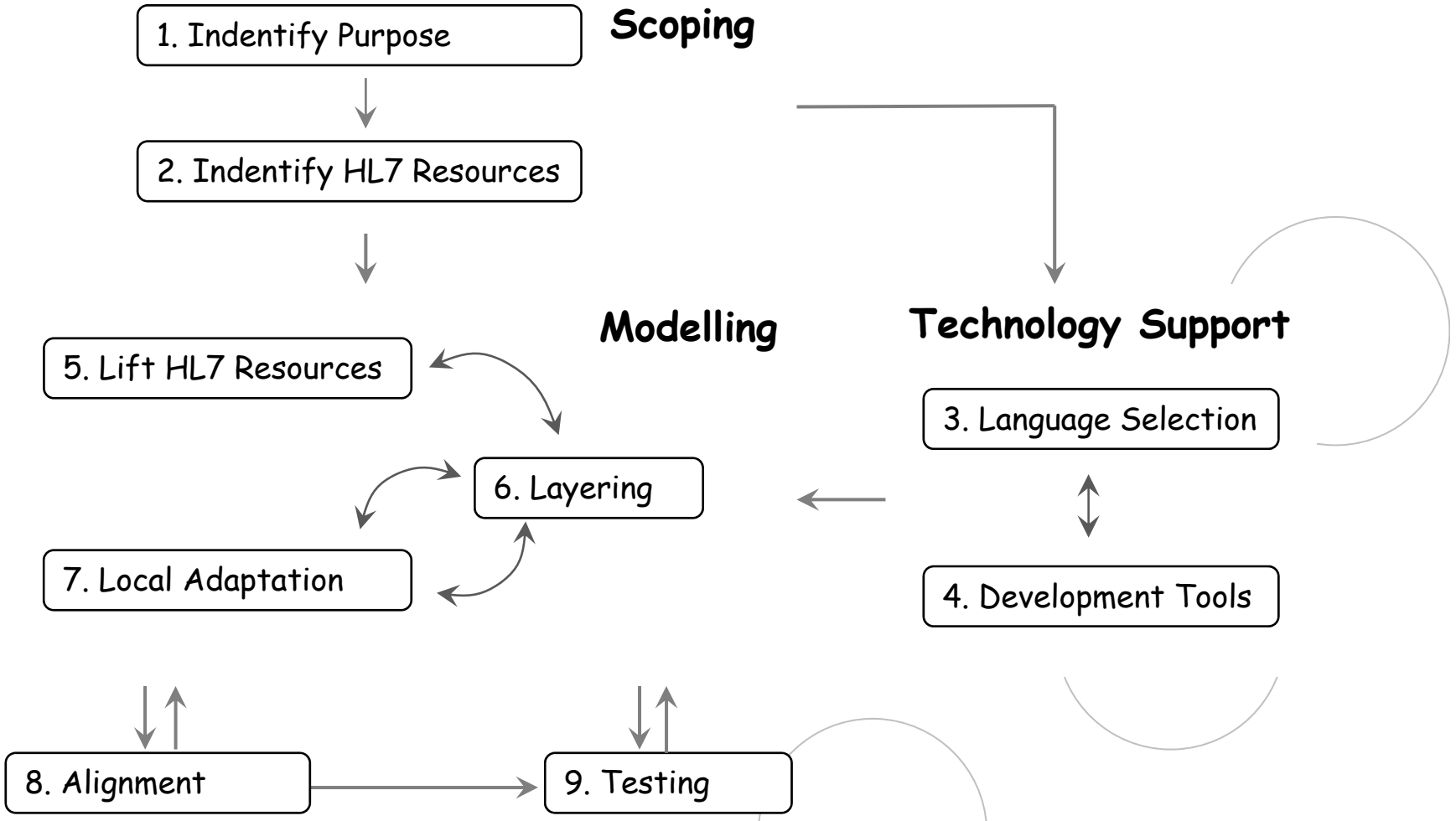
- Features Indentified
  - *Reusability* of non-ontological structured resources
  - *Layering* of ontologies
  - *Local adaptation* of ontologies

	Enterprise Ontology	METHONTOLOGY	On-To -Knowledge	DILIGENT
Reusability	+/-	+/-	+/-	+/-
Layering	-	-	-	+/-
Local Adaptation	-	-	-	+

PPEPR Methodology



PPEPR Methodology



Modelling: Lifting HL7 Resources

Language Transformation: A Hard problem

	XML Schema	Ontology
Data type	(1) Supports **large number of data types**	(1) RDFS/OWL 1 has limited support, thanks to OWL 2 for **extended data types support**
Structure	(1) **Nested** data structure (2) **Tree** structure (top element is root) (3) **Sequence** to describe element order	(1) **Concept composition** is through properties (2) **Graph** based (Any concept could be root) (3) **No ordering** of concepts
Relation	(1) Inheritance through **Type** and **Extension** (2) No Support	(1) **Multiple Inheritance** (2) Inheritance on properties and **logical implications** (symmetric, Transitive, etc.)

Transformation Rules

- MIF2OWL
- XSD2OWL

	HL7 MIF	OWL
element|attribute	attribute	ObjectProperty|DataProperty
element@substitutionGroup	StaticModel.association| StaticModel.attribute	SubPropertyOf
element@type	type	Range
complexType|group|attributeGroup	class | containedClass	Class
extension@base|restriction@base *union@memberTypes* *attribute@classCode type=Class*	childClass	SubClassOf
@maxOccurs @minOccurs	maximumMultiplicity| minimumMultiplicity	max|min
Annotation@appinfo *hl7:LongName|hl7:Type*	otherAnnotation | appInfo	Annotations@label|comment

Example

```
<xs:simpleType name="ActClassObservation">
   <xs:annotation>
     <xs:documentation>specDomain: S11529 (C-0-T11527-S13856-S11529-cpt)</xs:documentation>
   </xs:annotation>
  <xs:union memberTypes="ActCondition  ActClinicalTrial   ActSpecimenObservation  ActGenomicObservation ">
 </xs:union>
</xs:simpleType>
```

```
<xsl:for-each select="xsd:union[@memberTypes and parent::xsd:simpleType] |
xsd:simpleContent/xsd:union[@memberTypes and parent::xsd:simpleContent" ]

<xsl:if test="@memberTypes">
   <xsl:for-each select="tokenize(@memberTypes, '\s')">
       Class: <xsl:value-of select="." />
      SubClassOf:
    <xsl:value-of select="$currentClass"/>
</xsl:for-each>
```

```
Class: ActCondition SubClassOf: ActObservation
Class: ActClinicalTrial SubClassOf: ActObservation
Class: ActSpecimenObservation SubClassOf: ActObservation
Class: ActGenomicObservation SubClassOf: ActObservation
```

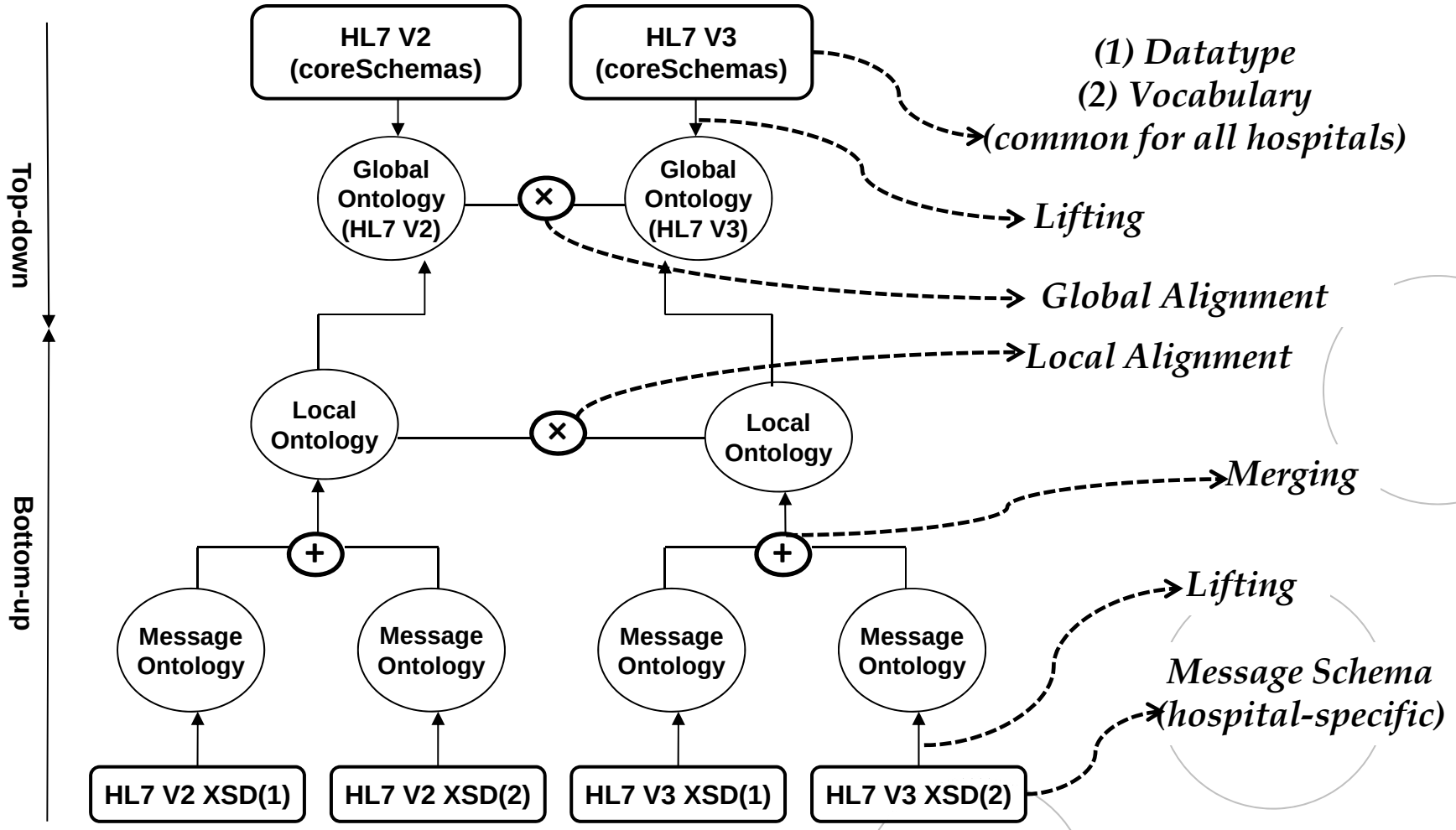
Example

```
<xs:complexType name="Patient">
 <xs:sequence>
    <xs:element maxOccurs="unbounded" minOccurs="1" name="id" type="II"/>
    <xs:element maxOccurs="1" minOccurs="1" name="name" type="EN"/>
    <xs:element maxOccurs="1" minOccurs="1" name="administrativeGenderCode" type="CE"/>
    <xs:element maxOccurs="1" minOccurs="1" name="birthTime" type="TS"/>
    <xs:element maxOccurs="unbounded" minOccurs="1" name="addr" type="AD"/>
     .....
</xs:sequence>
   <xs:attribute fixed="PSN" name="classCode" type="EntityPerson" use="optional"/>
</xs:complexType>
```

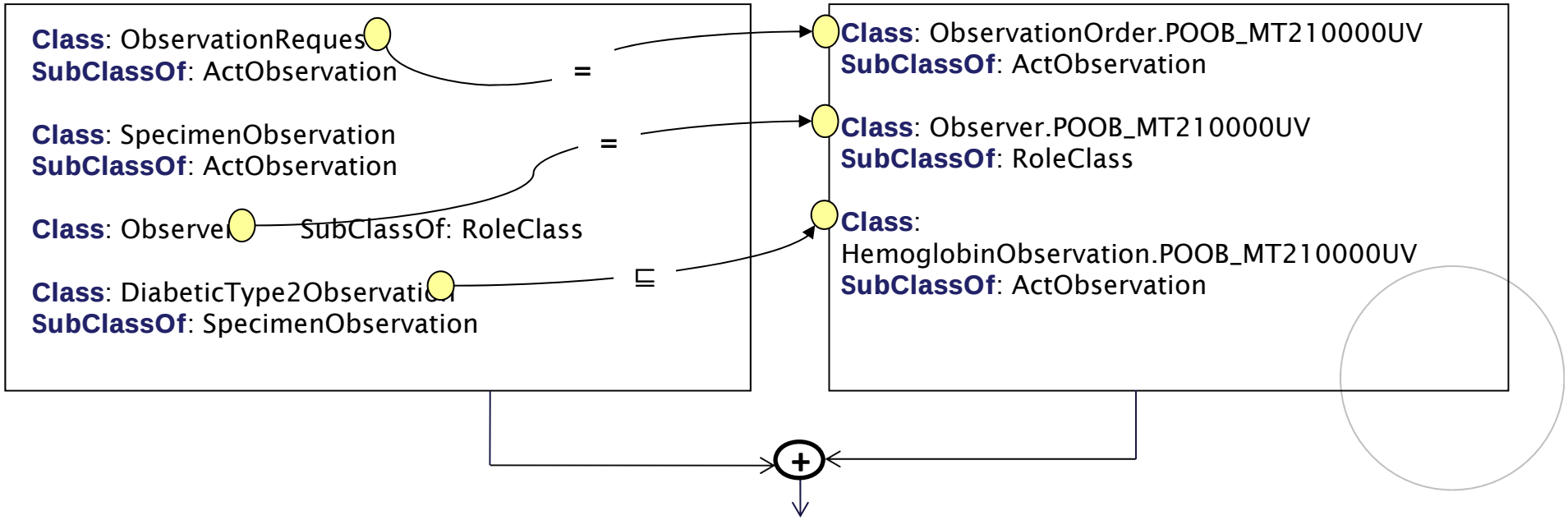
```
Class:   <xsl:value-of select="$currentClass"/>
<xsl:for-each select="xsd:attribute[@name="classCode"]
  <xsl:if test="@name='classCode'">
   SubClassOf: <xsl:value-of select="@type"/>
</xsl:for-each>
```

Class: Patient **SubClassOf**: EntityPerson
ObjectProperty: id **Domain**: Person **Range**: II
ObjectProperty: name **Domain**: Person **Range**: EN
ObjectProperty: administrativeGenderCode **Domain**: Person **Range**: CE
ObjectProperty: birthTime **Domain**: Person **Range**: TS
ObjectProperty: addr **Domain**: Person **Range**: AD

Layering of Ontologies



Local Ontology: Merging Local Ontologies



Class: ObservationRequest **SubClassOf**: ActObservation

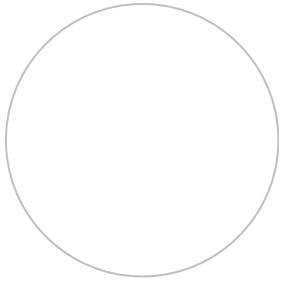
Class: SpecimenObservation **SubClassOf**: ActObservation

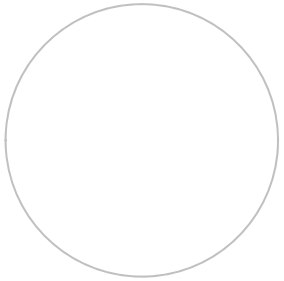
Class: Observer **SubClassOf**: RoleClass

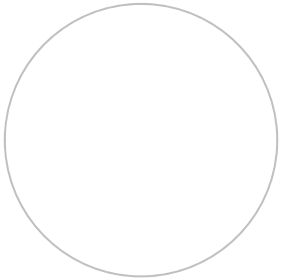
Class: HemoglobinObservation.POOB_MT210000UV **SubClassOf**: ActObservation

Class: DiabeticType2Observation **SubClassOf**: SpecimenObservation
HemoglobinObservation.POOB_MT210000UV

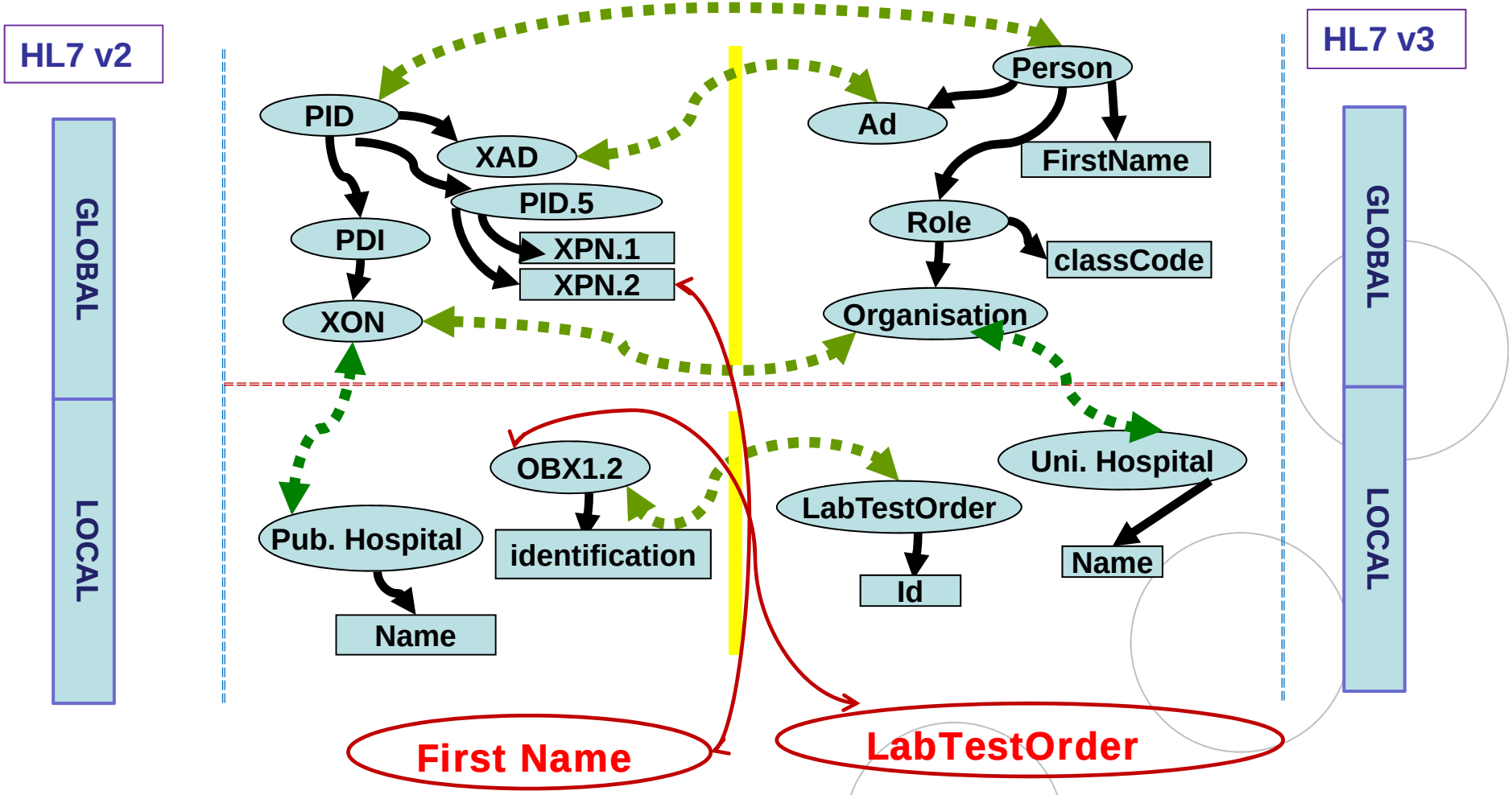
Aligning HL7 ontologies







Alignment: HL7 Global and Local Ontologies



Alignment: Example

Version 3	Version 3
Class: ObservationRequest **SubClassOf**: ActObservation **Class**: SpecimenObservation **SubClassOf**: ActObservation **Class**: Observer **SubClassOf**: RoleClass **Class**: DiabeticType2Observation **SubClassOf**: SpecimenObservation	**Class**: ObservationOrder.POOB_MT210000UV **SubClassOf**: ActObservation **Class**: Observer.POOB_MT210000UV **SubClassOf**: RoleClass **Class**: HemoglobinObservation.POOB_MT210000UV **SubClassOf**: ActObservation

Version 3	Version 2
Class: AD **SubClassOf**: ANY **ObjectProperty**: streetAddressLine **Domain**: AD **Range**: Adxp.country **ObjectProperty**: state **Domain**: AD **Range**: Adxp.state **ObjectProperty**: city **Domain**: AD **Range**: Adxp.city	Class: AD **ObjectProperty**: AD.1 **Domain**: AD **Range**: AD.1.CONTENT **ObjectProperty**: AD.2 **Domain**: AD **Range**: AD.2.CONTENT **ObjectProperty**: AD.3 **Domain**: AD **Range**: AD.3.CONTENT

HL7 Annotation

```
<xsd:complexType name="AD.3.CONTENT">
    <xsd:annotation>
        <xsd:appinfo>
            <hl7:Type>ST</hl7:Type>
             <hl7:LongName>City</hl7:LongName>
           </xsd:appinfo>
</xsd:annotation>
```

Ontology Alignment Tools

Method/Tool	HL7 (V3 –V3) precision-recall (Local Ontologies)	HL7 (V2-V3) precision-recall (Global/Local Ontologies)	Threshold Value
Falcon-AO	70%(p)-60%(r) 70%(p)-50%(r) 70%(p)-50%(r)	30%(p)-30%(r) 30%(p)-20%(r) 30%(p)-20%(r)	0.1-0.4 0.4-0.7 0.7-1
H-Match	80%(p)-100%(r) 80%(p)-90%(r) 80%(p)-90%(r)	40%(p)-30%(r) 40%(p)-20%(r) 40%(p)-20%(r)	0.1-0.4 0.4-0.7 0.7-1
BLOOMS	90%(p)-40%(r) 90%(p)-30%(r) 90%(p)-30%(r)	90%(p)-20%(r) 90%(p)-10%(r) 90%(p)-10%(r)	0.1-0.4 0.4-0.7 0.7-1
RiMOM	60%(p)-100%(r) 70%(p)-90%(r) 70%(p)-90%(r)	40%(p)-40%(r) 40%(p)-40%(r) 30%(p)-20%(r)	0.1-0.4 0.4-0.7 0.7-1
AgreementMaker	70%(p)-100%(r) 70%(p)-90%(r) 70%(p)-90%(r)	40%(p)-50%(r) 40%(p)-50%(r) 30%(p)-20%(r)	0.1-0.4 0.4-0.7 0.7-1

Alignment: SPARQL Recipes

Class Matching

```
CONSTRUCT { ?v3 owl:equivalentClass ?v2 }
WHERE { ?v3 rdf:type owl:Class . ?v2 rdf:type owl:Class .
        ?v2  rdfs:label    ?LongName .
{FILTER regex(str(?v3), str(?LongName), ``i'')}}
```

Property Matching

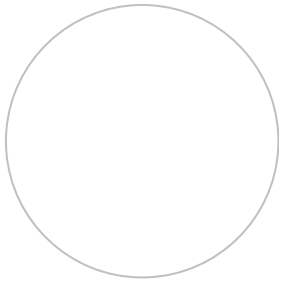
```
CONSTRUCT { ?v3 owl:equivalentProperty ?v2 }
WHERE { ?v3 rdf:type owl:ObjectProperty .
        ?v2 rdf:type owl:ObjectProperty .
        ?v2 rdfs:range ?v2range .
        ?v3 rdfs:range ?v3range .
        ?v2 rdfs:domain ?v2domain .
        ?v3 rdfs:domain ?v3domain .
        ?v2range owl:equivalentClass ?v3range .
        ?v2domain owl:equivalentClass ?v3domain };
```

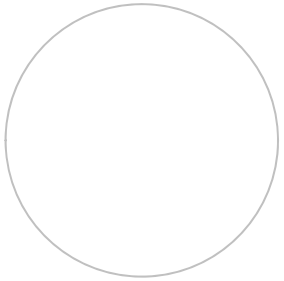
Alignment: SPARQL Recipes

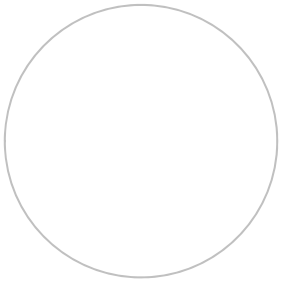
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H-Match	80%(p)-100%(r) 80%(p)-90%(r) 80%(p)-90%(r)	40%(p)-30%(r) 40%(p)-20%(r) 40%(p)-20%(r)	0.1-0.4 0.4-0.7 0.7-1
BLOOMS	90%(p)-40%(r) 90%(p)-30%(r) 90%(p)-30%(r)	90%(p)-20%(r) 90%(p)-10%(r) 90%(p)-10%(r)	0.1-0.4 0.4-0.7 0.7-1
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AgreementMaker	70%(p)-100%(r) 70%(p)-90%(r) 70%(p)-90%(r)	40%(p)-50%(r) 40%(p)-50%(r) 30%(p)-20%(r)	0.1-0.4 0.4-0.7 0.7-1
SPARQL Recipes	**80%(p)-90%(r)**	**50%(p)-60%(r)**	**NA**

Extend **alignment tools** (AgreementMaker, RiMOM) by including **domain-specific thematic structures** instead of general information structures like WordNet, Wikipedia, DBpedia

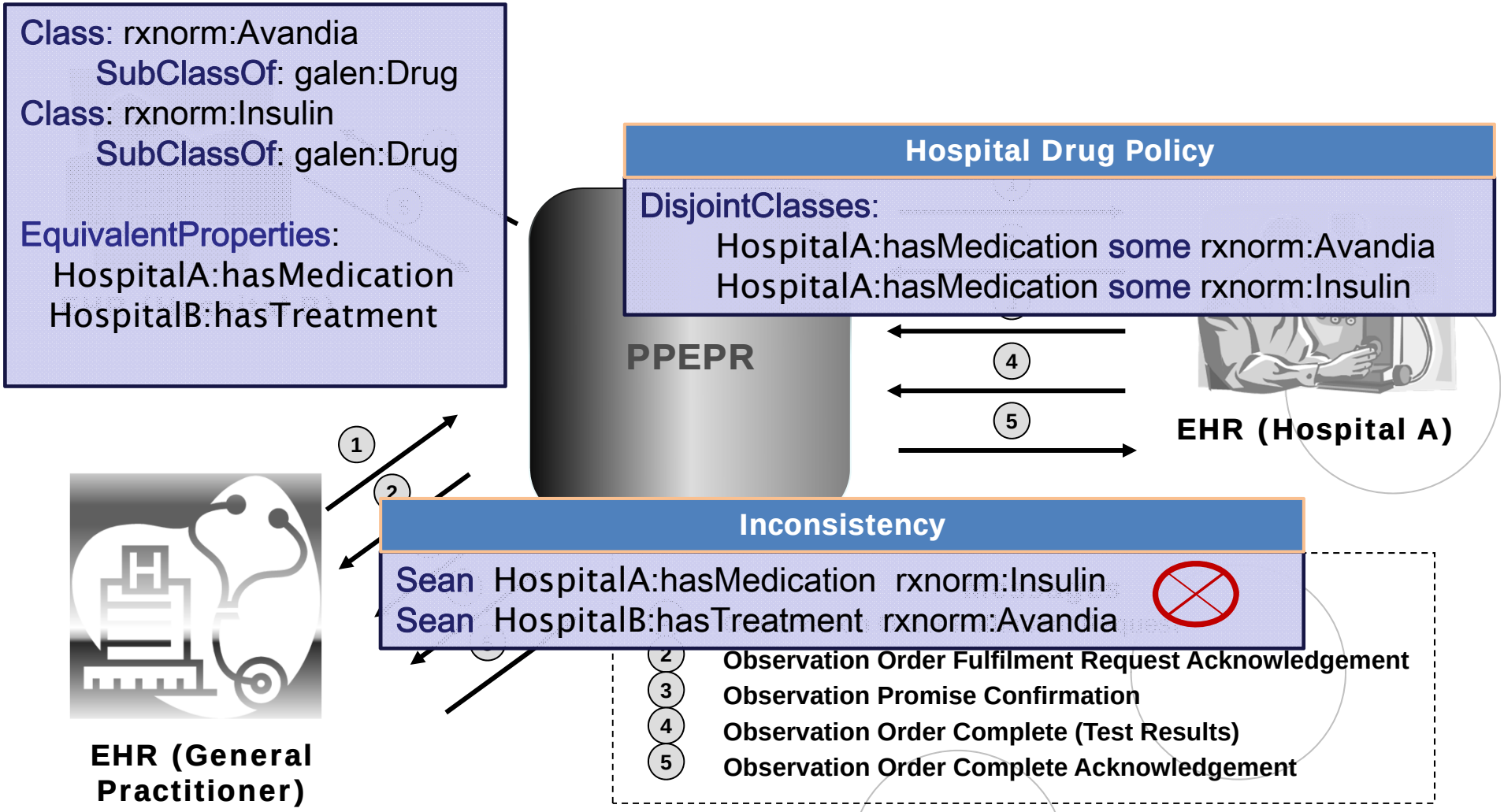
Context, Modularity and Local Policies







Example Scenario



Where is the Fault ?

- Ontologies are
  - *„Specification of a conceptualization."* — *Tom Gruber, 1993*
  - Agreed between groups with explicit semantics. — OWL Semantics, W3C, 2004
  - Monotonic and make Open World Assumption (OWA). — OWL Semantics, W3C, 2004
  - Good at Description of Reality and their mappings.
- Ontology are not
  - Model of local and context-specific information
  - Model of time-dependent information
  - Model of context-specific constraints (e.g., policy, preferences) and validation

State-OF-The-Art -1 : Formal Approaches

- We did investigation for support of five features
  - Context-awareness (CA)
  - Modularity (M)
  - Profile and policy management (P & PM)
  - Correspondence expressiveness (CE)
  - Robustness to heterogeneity (RH)

Considered Approaches:

- Standard DL: Web Ontology Language (OWL)
  - No **localised** or **contextualised** semantics
  - Reusability or knowledge integration is **limited to *owl:imports***
- Context-Extensions of DLs : Distributed Description Logic (DDL)
  - Packet Description Logic (PDL)
  - Integrated Distributed Description Logic (iDDL)
  - E-connection
- DL+Constraints/Rules
  - DL+DL-Safe Rules
  - Database-Style Integrity Constraints (IC) within OWL (OWL/IC in Pellet)
- Rule-based
  - Modular Web Rule Bases
- Query-Based
  - Query-Translation
- Repairing and Reasoning with Inconsistencies (DeLP)

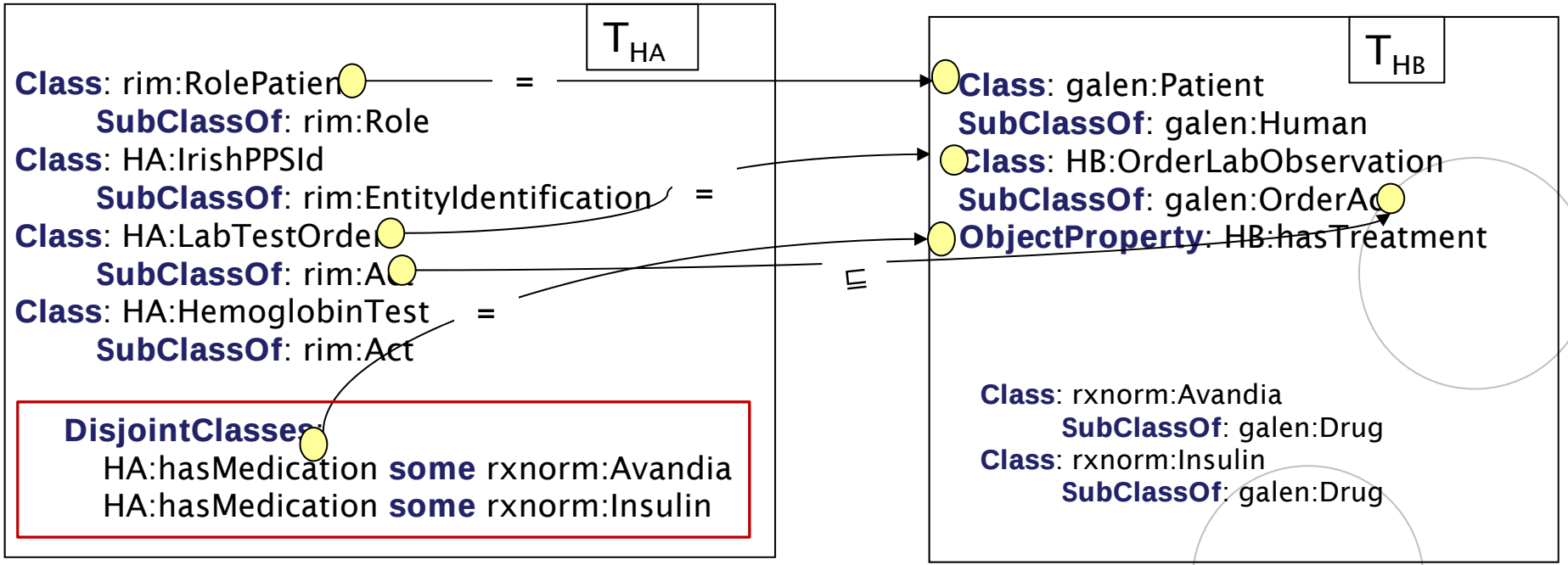
NONE OF THEM ADDRESSES ALL FEATURES

State-of-the-Art-2 (RDF)

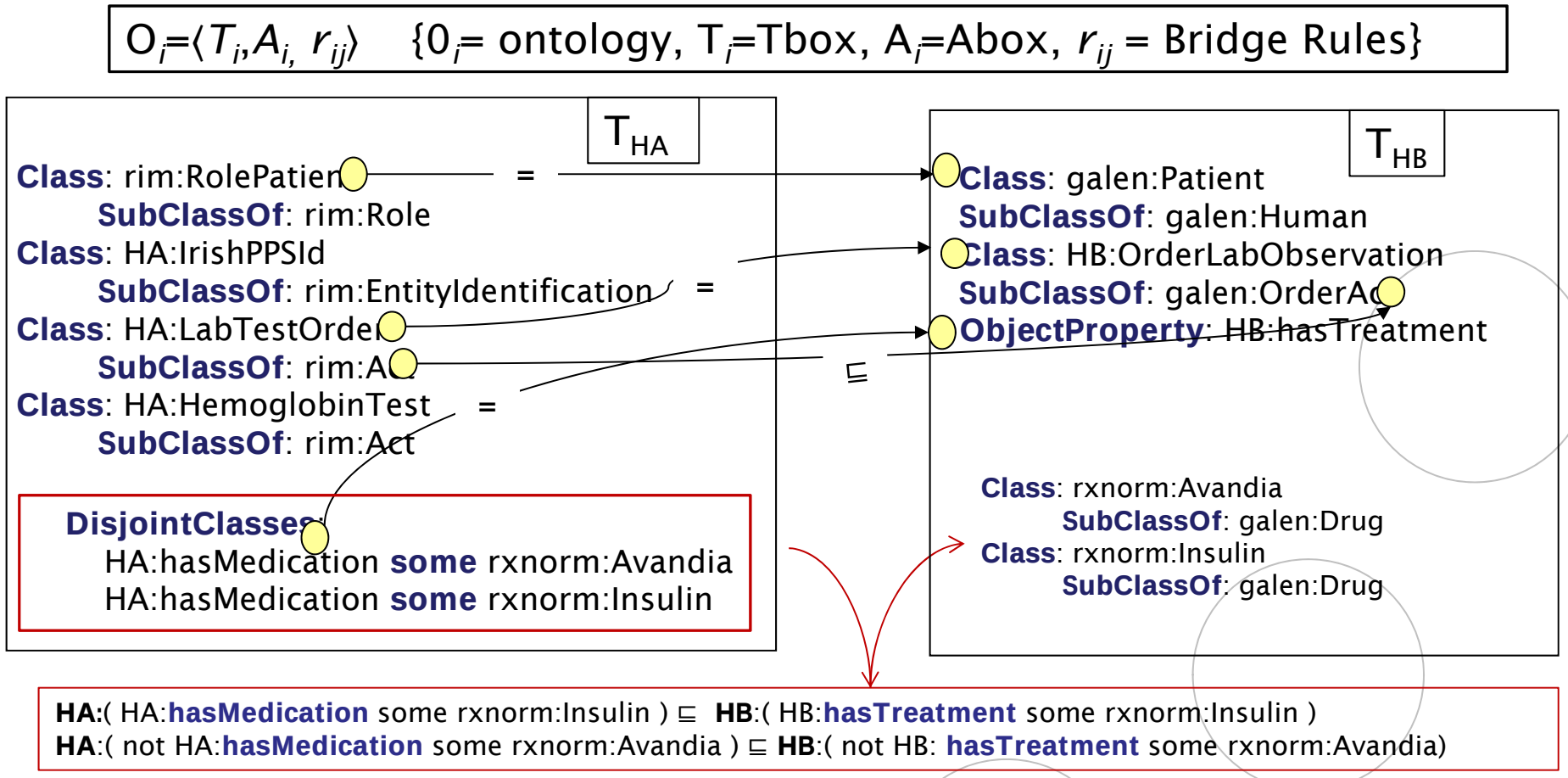
- Resource Description Framework (RDF)
  - RDF is an assertional logic (antecedent or premises is always true), where each triple expresses a simple proposition. *[W3C RDF Semantics document]*
    - In result, triple (s p o) represent facts, notion of "universal truth".
    - RDF triples are context-free
- Reification
  - N statements about a statement
  - Good for making statements about provenance
  - NO coupling with the truth of the triple that has been reified
  - Cannot relate the truth of a triple in one context (graph) to another
- Named Graphs
  - Assigned an ID (URI) to each graph
  - Good for making statements about provenance
  - Associate named graphs with triples
    - Triples become quadruples
    - Fourth element is the URI of the named graph (origin)
  - Similar to Reification for the "truth of a triple"
- N3-Context
  - Similar to Reification as far as "truth of a triple" is concerned

Standard Semantics : OWL

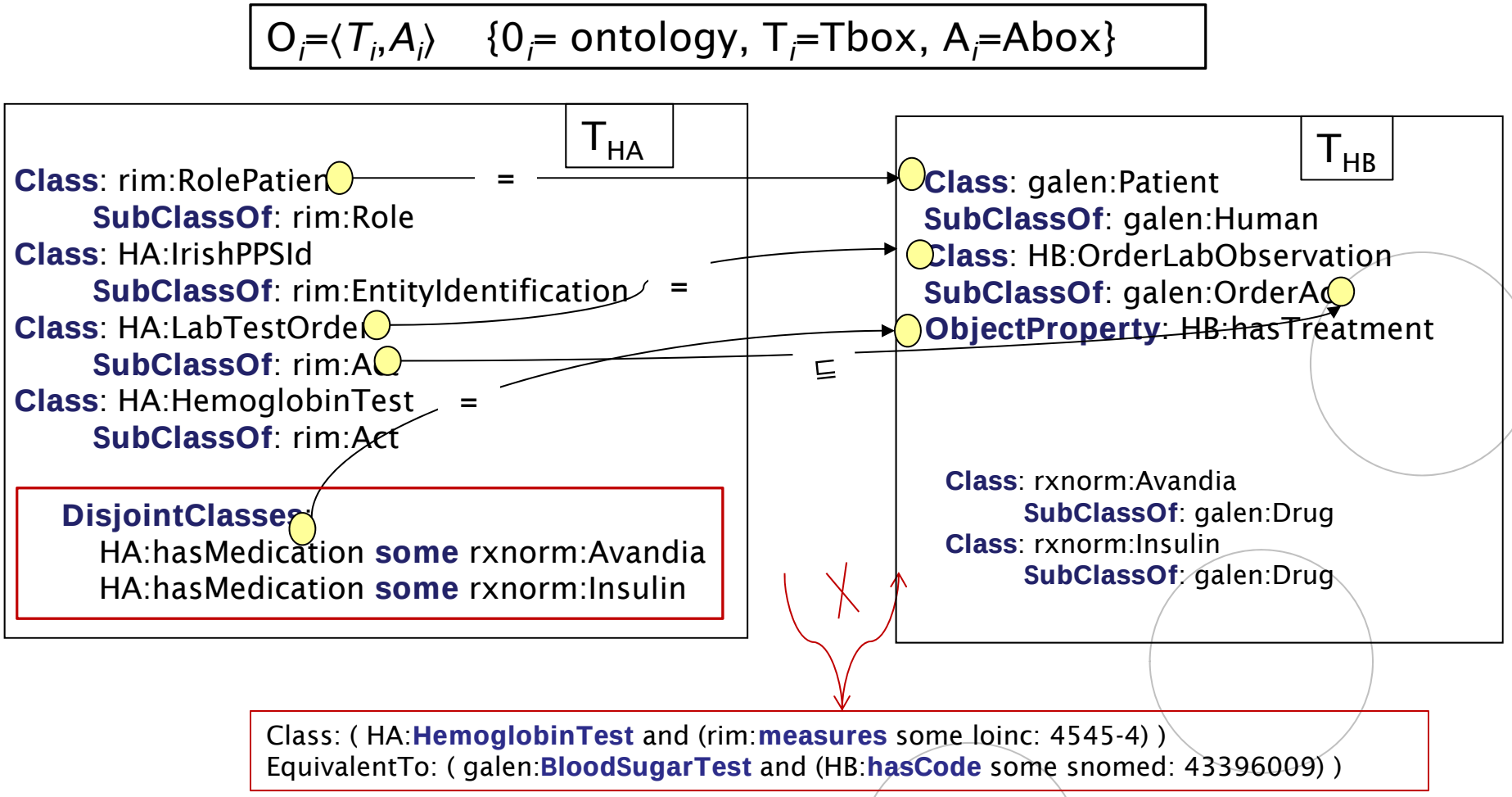
O=⟨*T*,*A*⟩ {0= ontology, T=Tbox, A=Abox}



Distributed Description Logic (DDL)

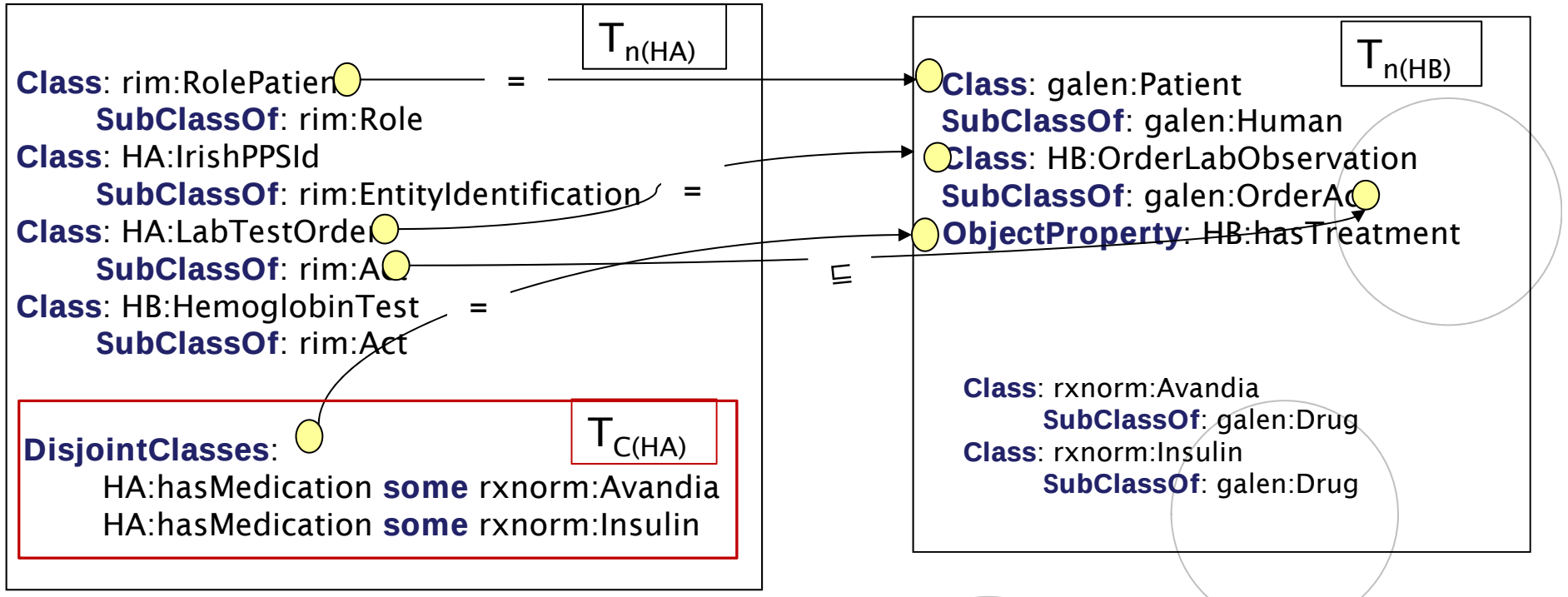


Packet Description Logic (PDL)



Database Style - IC

O=⟨T_n, $T_{C,}$ A⟩ {0= ontology, T_n =Normal Tbox,T_C = *Constraint Tbox*, A=Abox}



Feature Comparisons

	Context-awareness	Modularity	Profile & Policy Management
DL/OWL	-	-/+	-
DDL/C-OWL	+	+	-
P-DL	+	+	-
DDL Revisited	+	+	-
IDDL	+	+	-
E-connection	+	+	-
RDFS-C (Guha's)	+	-/+	-
Query-based	-/+	-	-
Modular Rule bases	+	+	-/+
OWL/IC	-	-/+	-/+
DeLP/Paraconsistent	-	-	-/+

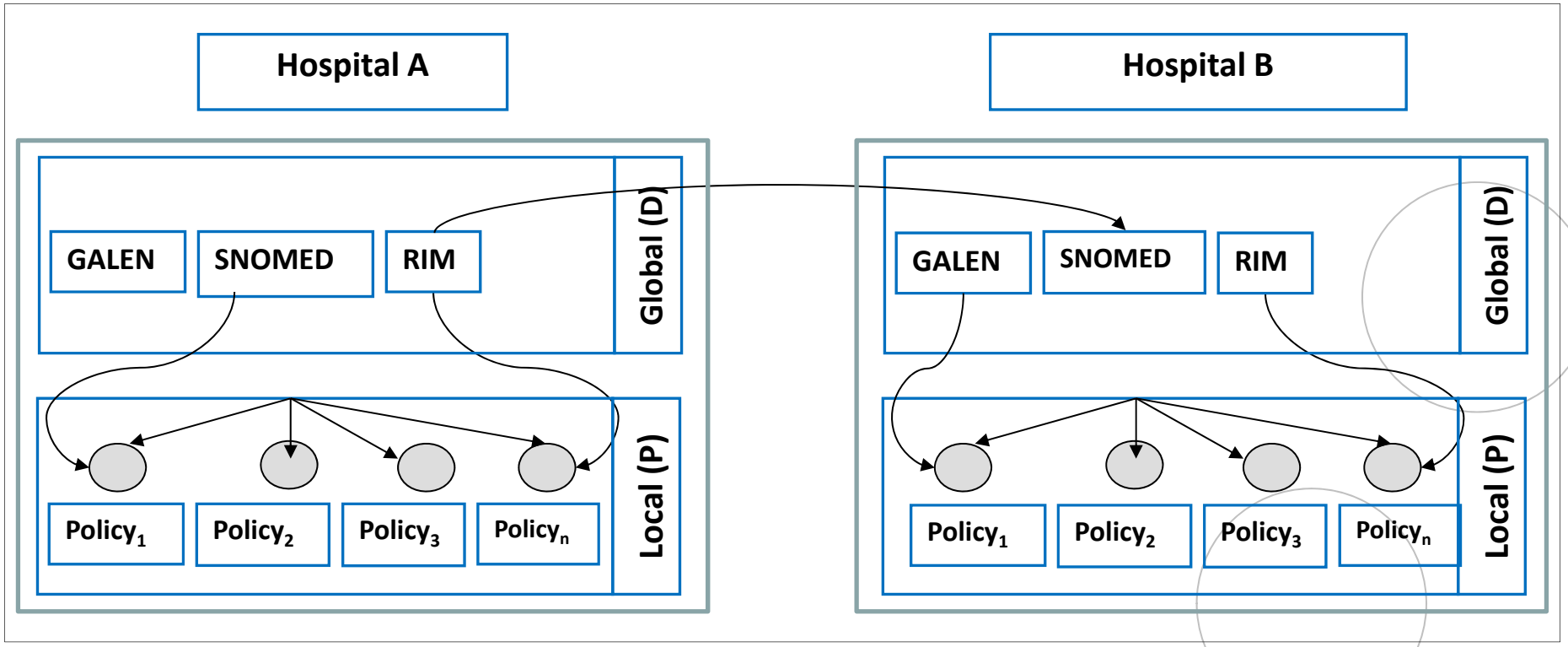
Feature Comparisons

	C.A	M.	P. & PM	C.E	R.H
DL/OWL	-	-/+	-	Good	Very Limited
DDL/C-OWL	+	+	-	Very Good	Good
P-DL	+	+	-	Very Limited	Limited
DDL Revisited	+	+	-	Very Good	Medium
IDDL	+	+	-	Good	Very Good
E-connection	+	+	-	Medium	Excellent
RDFS-C	+	-/+	-	Good	Good
DeLP/Paraconsistent	-	-	-/+	Good	Good
Query-based	-/+	-	-	Very Good	Very Good
Modular Rule bases	+	+	-/+	Limited	Limited
OWL/IC	-	-/+	-/+	Good	Limited

C.A: Context-awareness, **M**: Modularity, **P & PM**: Profile and policy management,
CE: Correspondence expressiveness, **RH**: Robustness to heterogeneity

Envisioned Situation

- ***Context & Policy aware ontological model and reasoning***



Summary

- An ontology is good at the **top-down modeling** of a domain
  - reduces the bilateral correspondences between healthcare applications
  - delegates the majority of mediation to the central integration location
- An ontology provides an **executable** (comparing to HL7 UML model) semantics and consistent model
- The Semantic Web layer cake allows to engage information model, schema, and instances under a **single framework**. In HL7 they are represented in three isolated layers.
- An **automated ontology alignment** is a great support for the domain experts comparing manual syntactic alignment
- An ontology for the healthcare domain eases harmonising **Medical, Life Sciences, and Pharma** domains
  - Prominent vocabularies are already available as ontologies (SNOMED, OBI, EFO, RXNORM, Disease Ontology, Cell Type Ontology, etc.)
- An ontology has limitations in representing
  - **Contextual** and **modular** information
  - **Policy-based** information

Things cooking at the moment !

HL7 FHIR - OWL HL7 FHIR - RDF



http://www.hl7.org/implement/standards/fhir/

Thank you

Dr. Ratnesh Sahay
Semantics in e-Health and Life Sciences (SeLS)
Insight Centre for Data Analytics
NUI Galway, The DERI building
IDA Business Park, Lower Dangan
Galway, IRELAND
Tel: + 353 91 495253
Fax: + 353 91 495541
Web: http://www.ratneshsahay.org/



